

MYOLOGY

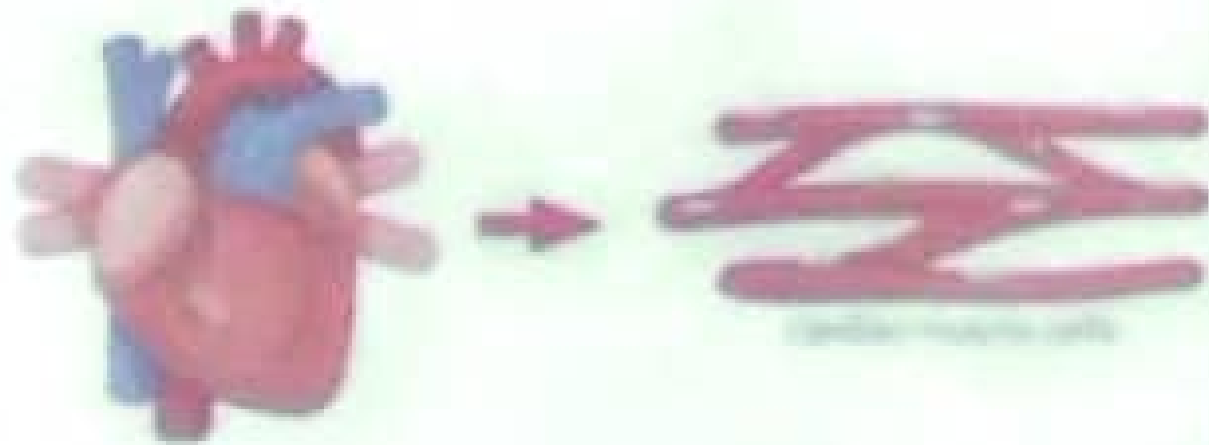
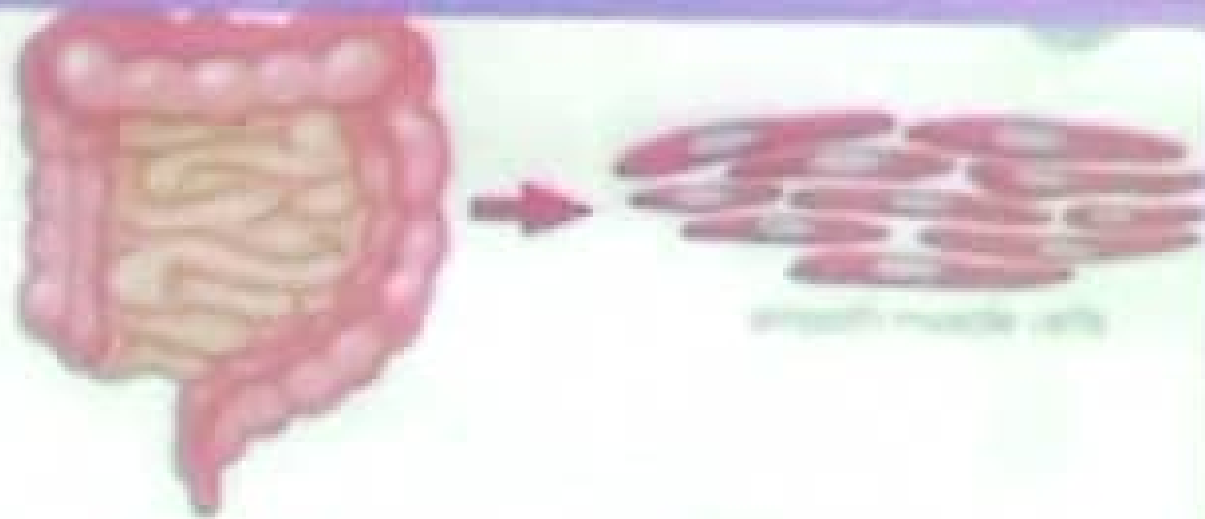
General Myology

■ **Myo:** muscle **ology:** science

Def.: science deals with muscles

the active part of locomotor system which has ability of contraction



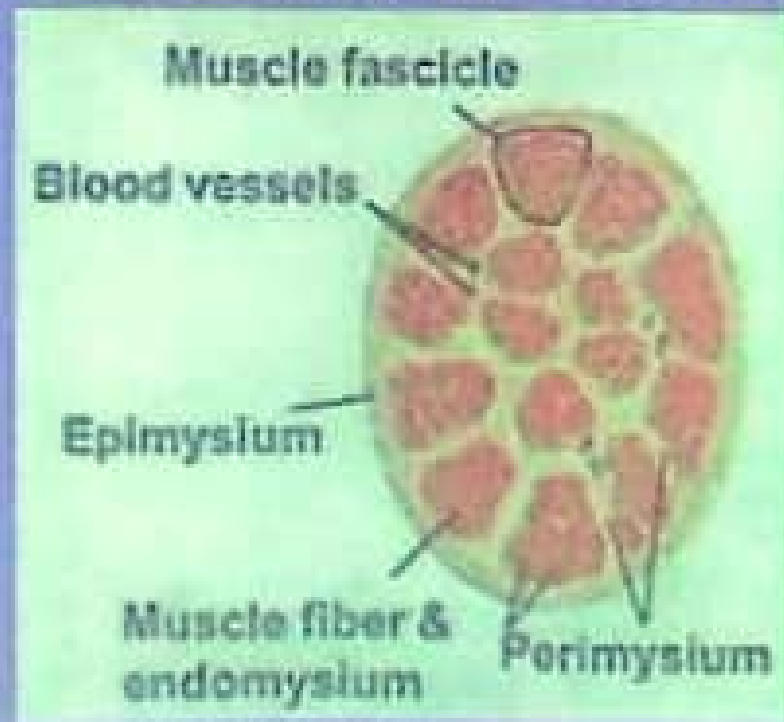


Muscle tissue classified morphology & function into

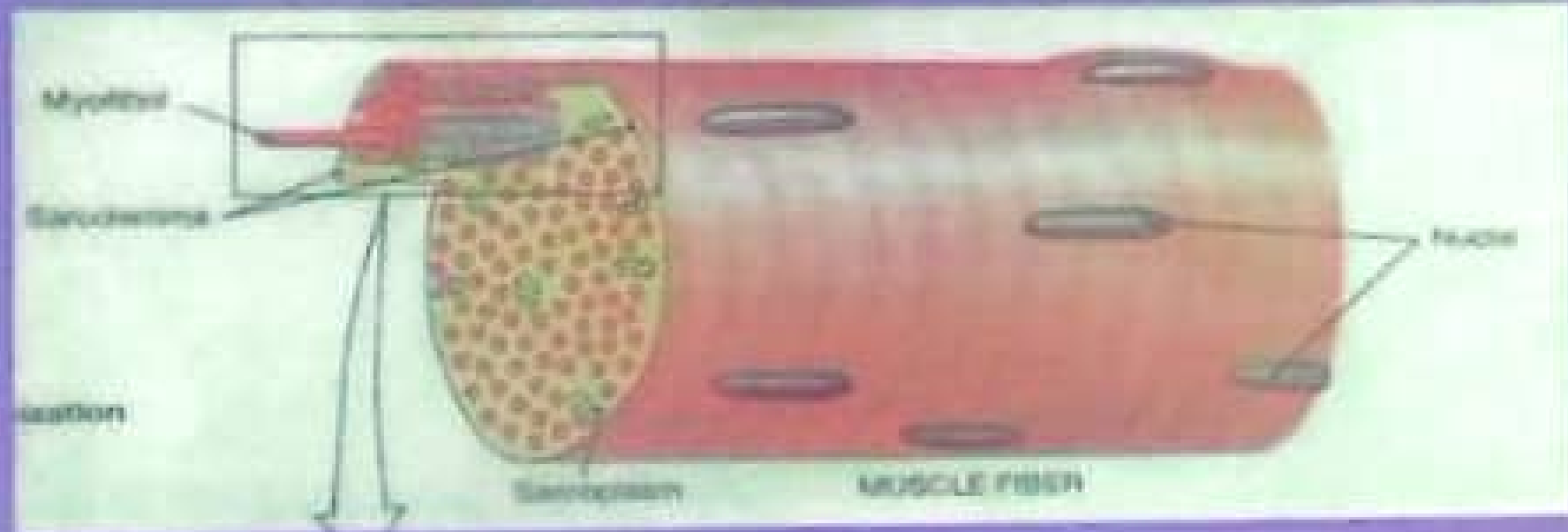
Smooth	Cardiac	Skeletal
➤ Non-striated ➤ Involuntary ☐ Autonomic nerves	➤ striated ➤ Involuntary ☐ Autonomic nerves	➤ striated ➤ Voluntary ☐ Somatic nerve
▪ The muscle fiber is spindle shaped ▪ large single and central nucleus.	bundles of cells each has single central nucleus. And branched	The muscle fibers are multinucleated cell with peripheral nuclei.
➤ wall of visceral organ and blood vessels	heart myocardium	Skeletal muscle

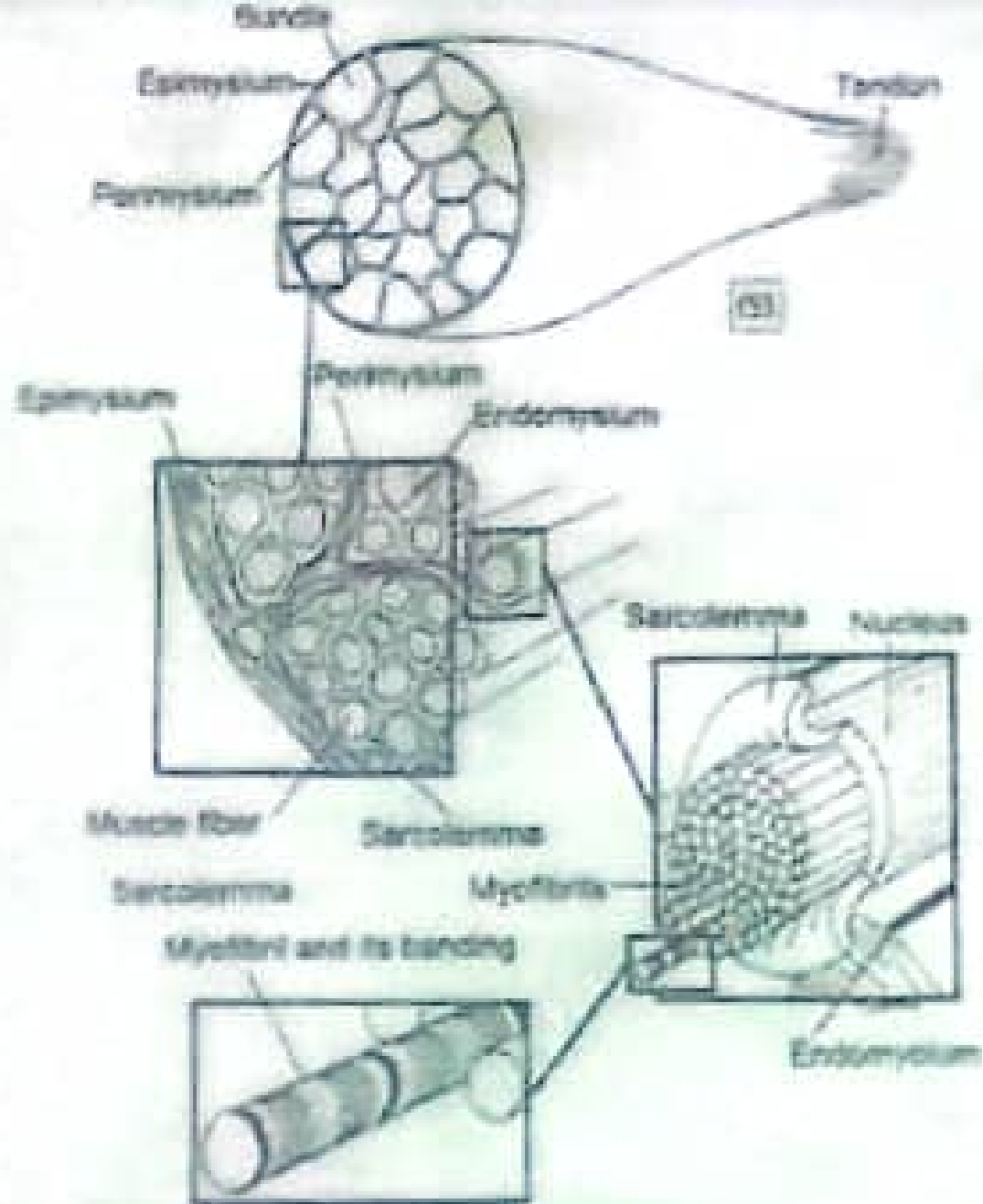
Architecture of skeletal muscle

- Muscle $\Rightarrow$ muscle bundles(fascicles) $\Rightarrow$ muscle fibers
- Connective tissue
- Epimysium $\Rightarrow$ perimysium $\Rightarrow$ endomysium
- perimysium + endomysium = tendon



Microscopic structures





Architecture of a skeletal muscle and its fibers.

Gross anatomy of skeletal muscle

2- skeletal muscle attachment

Bone (biceps brachii)

Cartilage (rhomboides)

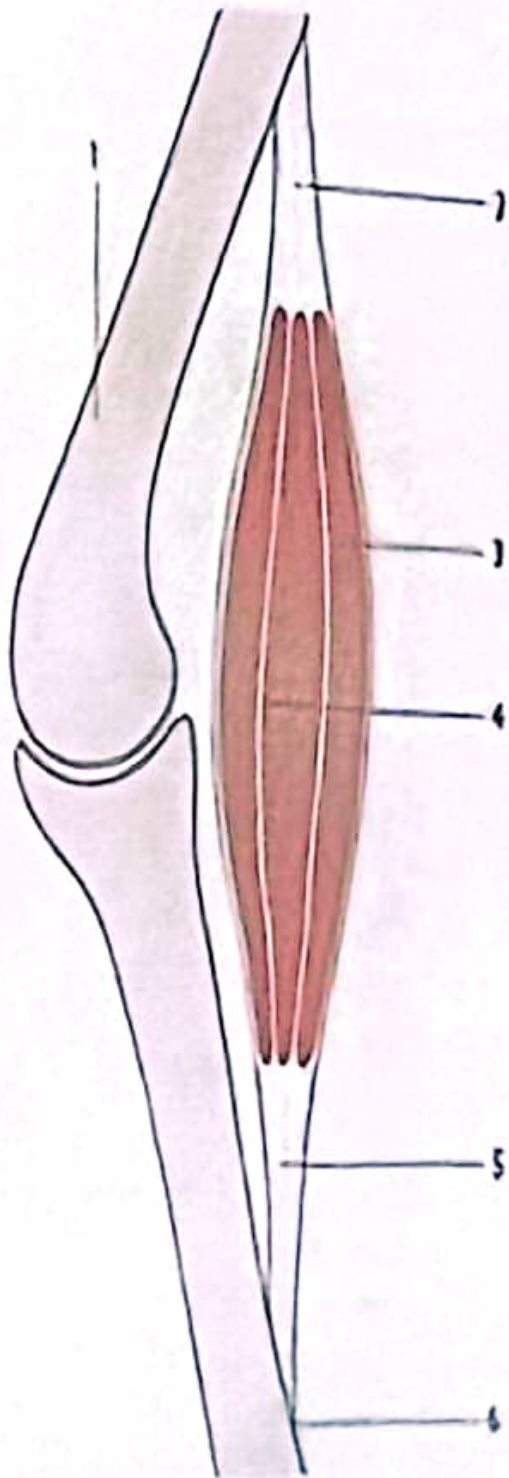
Deep fascia (indirectly to bone) (tensor fascia lata)

Skin and superficial fascia

Intermediate tendon (digastric muscle)

No attachment (no origin & no insertion, as orbicularis oris)

muscle

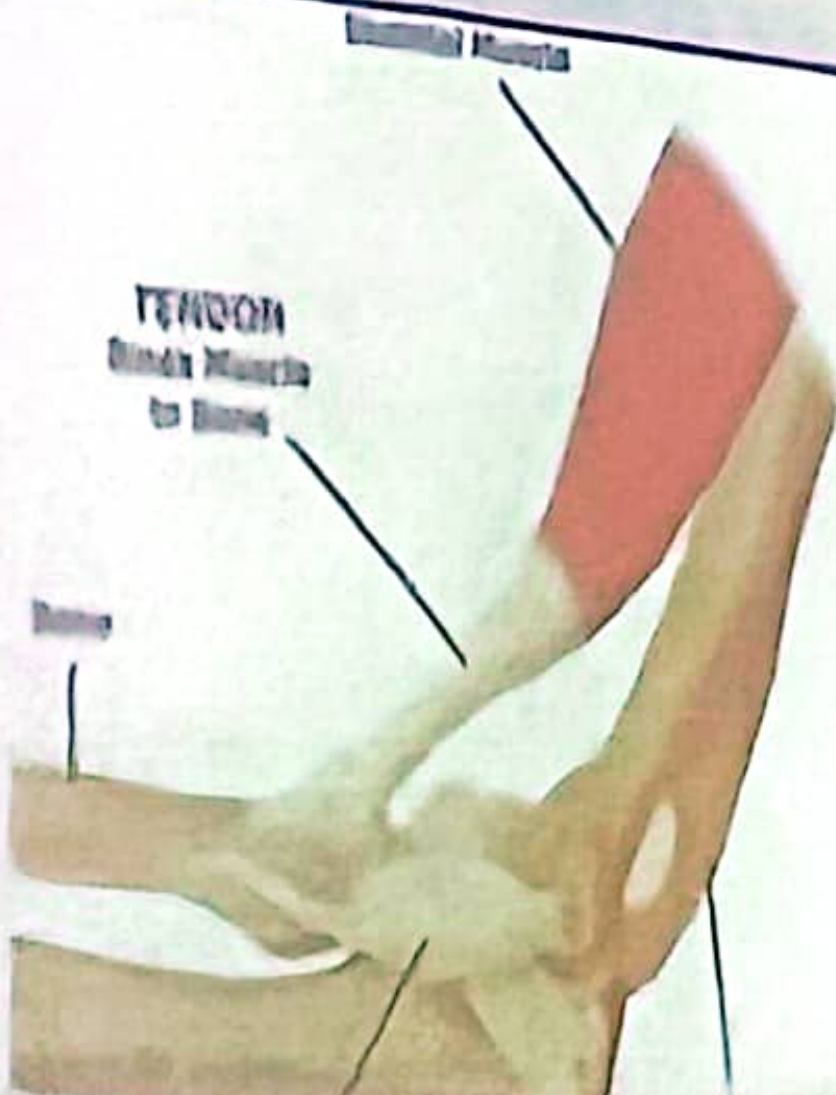


- 1 - Bone
- 2 - Origin
- 3 - Muscle belly
- 4 - Connective tissue between muscle fibres
- 5 - Tendon
- 6 - Insertion

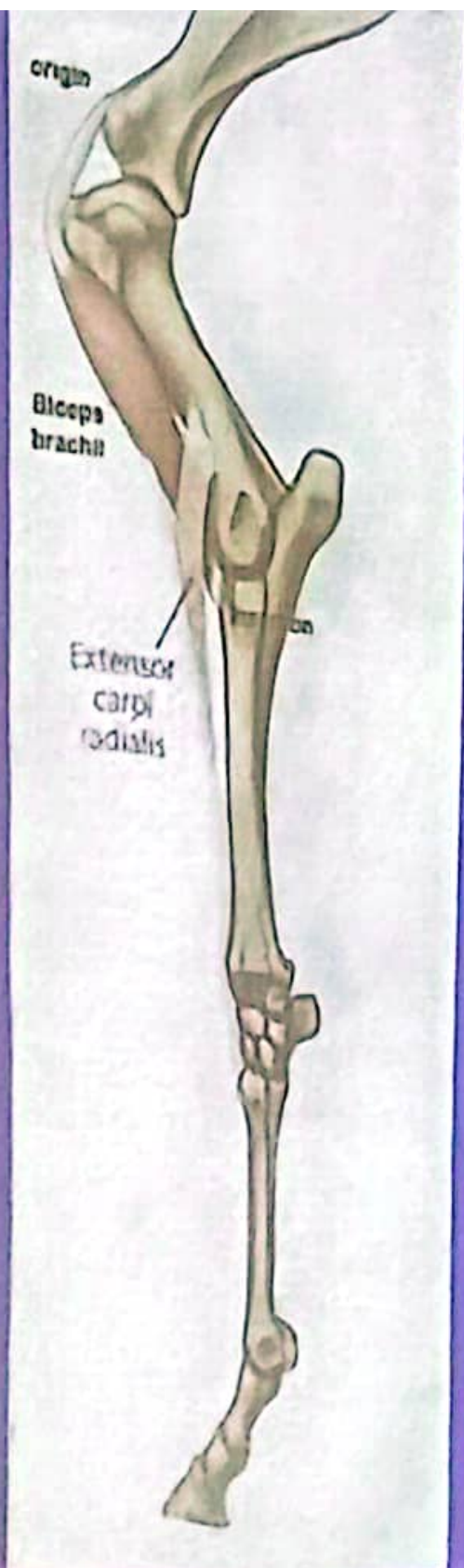
Aponeuroses.

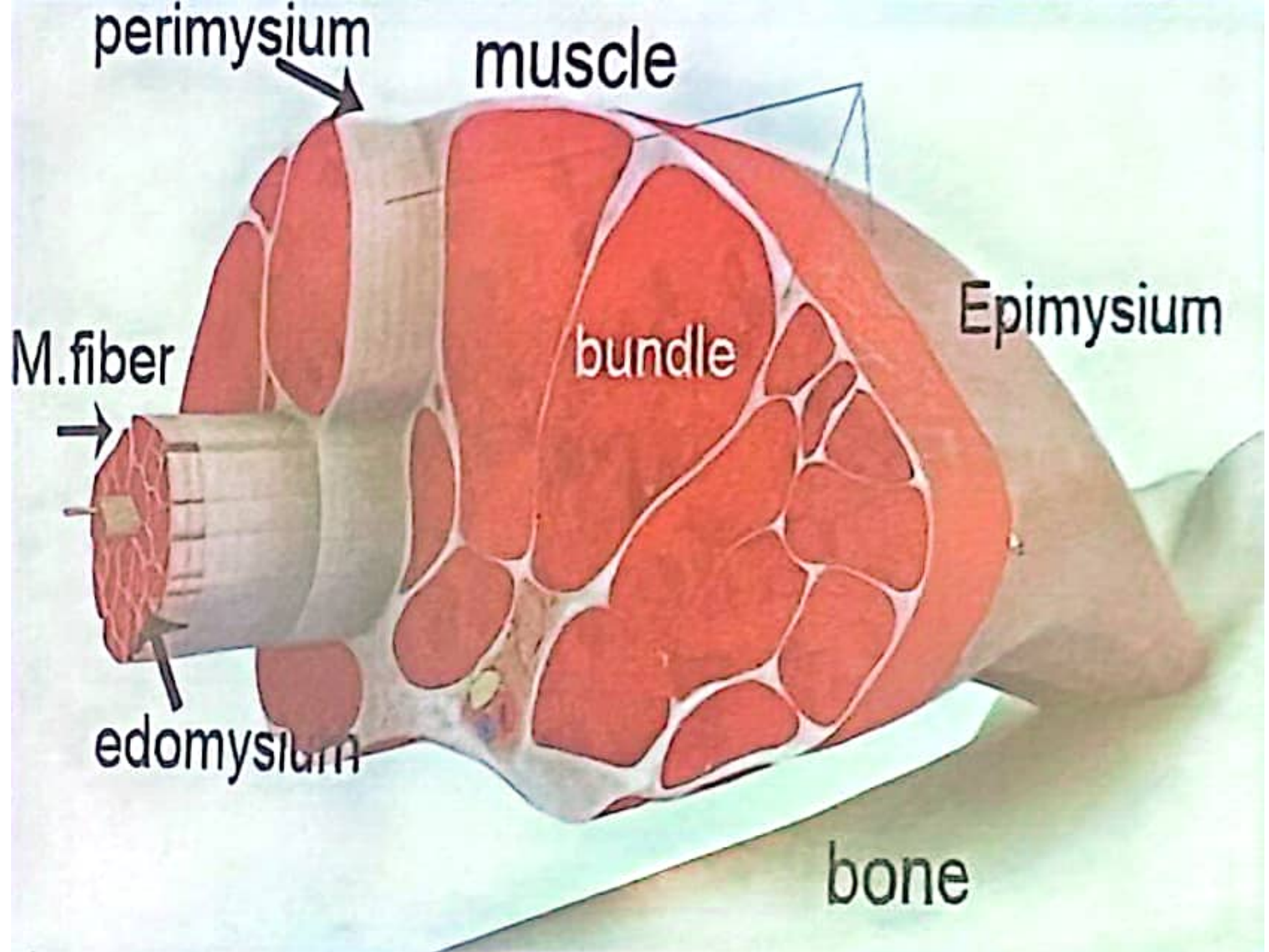
- Thin strong flat sheet of fibrous tissue
- Represent attachment of flat muscles
- Example : oblique muscles of abdomen.

TENDON
Quadratus Lumborum
to Ilium





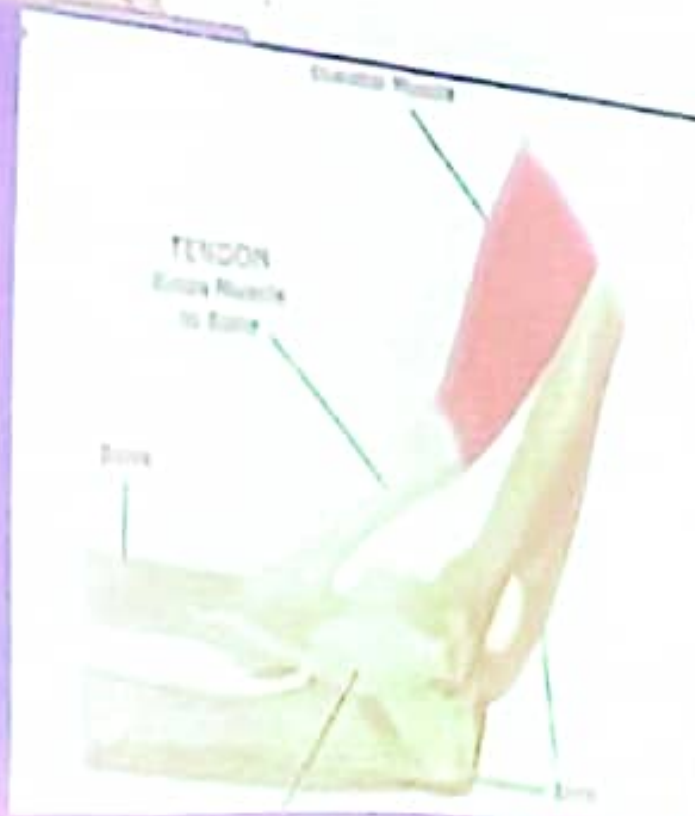






Aponeuroses.

- Thin strong **flat sheet of fibrous tissue**
- Represent attachment of flat muscles
- **Example** : oblique muscles of abdomen.



Gross anatomy of skeletal muscle

1- anatomical parts:

A- origin:

- the proximal attachment of muscle- no movement

B- insertion:

- Usually distal attachment- More movable

C- belly or gaster :

- The main balk of muscle between origin and insertion
- Tendon : band or cord like C.T
- Aponeurosis: thin flat and thin tendon for flat

Functional Grouping of Muscles

1. Flexor
2. Extensor
3. Adductor
4. Abductor
5. Constrictor
6. Dilator

Functional types of muscles

1. ***Prime mover (Agonist muscles)***

- which are responsible for initiation of a particular movement

2. ***Antagonist muscles***

- located on opposite side of joint
- have the opposite action to agonist

3. ***Synergist***

- assist in action of *agonists*
- not necessarily prime movers for the action

4. ***Fixators (stabilizers)***

- Fix the joint

Origin



fusiform



biceps



triceps



quadriceps

digas

Insertion



tendinous
intersection



multicaudate



serratus

2. According to origin

1. Uniceps .. Ex. Brachialis muscle.
2. Biceps .. Ex. Biceps brachii muscle.
3. Triceps .. Ex. Triceps brachii muscle.
4. Quadriceps .. Ex. Quadriceps femoris muscle.

1. Origin



fusiform



biceps



triceps

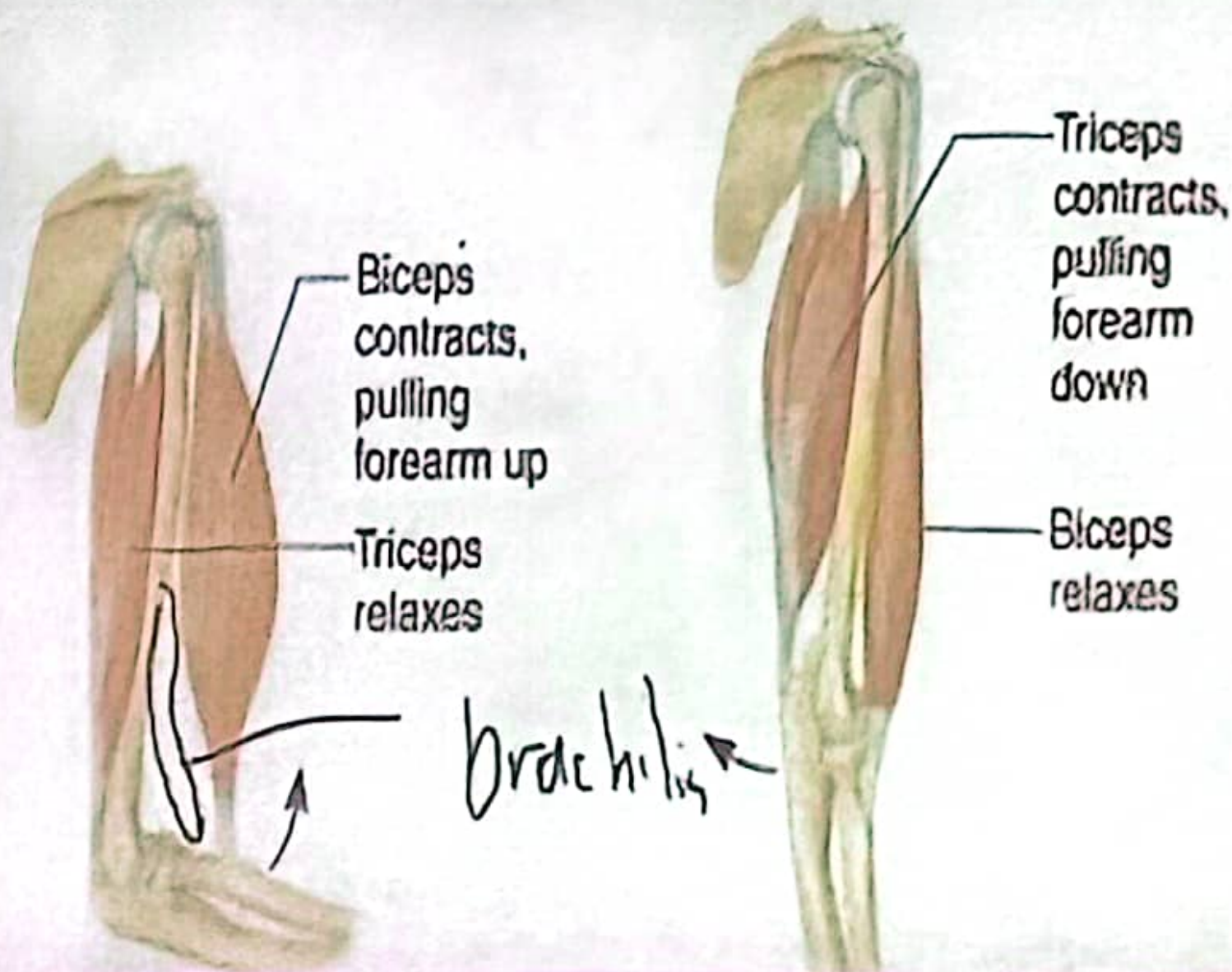


quadriceps



digastric

<i>Agonist</i>	Prime Mover	e.g. biceps
<i>Antagonist</i>	Opposite motion	e.g. triceps
<i>Synergist</i>	Aids agonist	e.g. brachialis



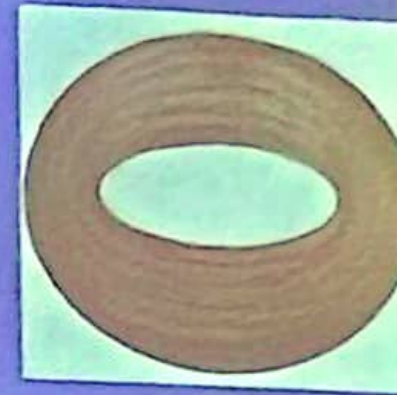
The skeletal muscles are named according to the following:

1. Attachment e.g. (sternothyrohyoid muscle.)
2. Position e.g. (supraspinatus muscle)
3. Shape e.g. (rhomboids muscle and rectus abdominis muscles.)
4. Structure e.g. (semimembranosus muscle.)
5. Action e.g. (extensor, flexor abductors and adductors muscles.)
6. Fibers arrangement e.g. (transverses abdominal muscle.)
7. Length e.g. (flexor digitorum longus and brevis.)

C- circular fibers

1- circular:

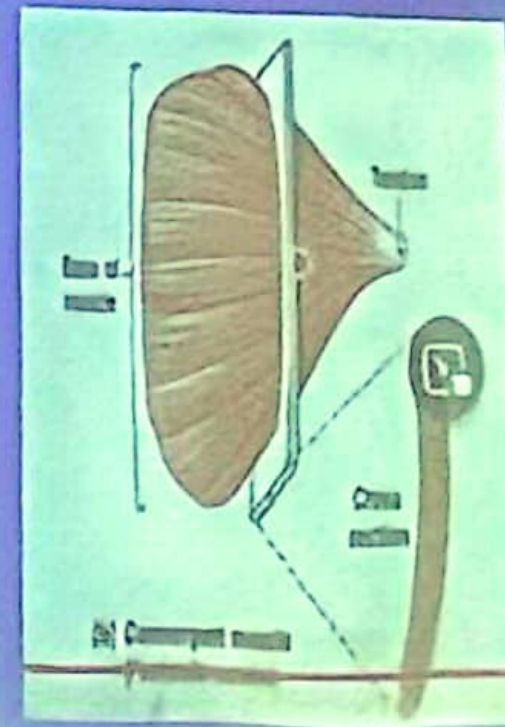
- no attachment, around orifices
- orbicularis oris (around mouth)



Circular

2- convergent:

- broad origin and pointed insertion
- Pectoralis major



Oblique fibers (pennation)
The fibers attached to tendon at oblique angle (feather like)

1- unipennate:

- Muscle fiber between tendon of origin and tendon of insertion pass in one direction
- Extensor digitorum longus M



Unipennate

2- bipennate:

- Muscle fiber between tendon of origin and tendon of insertion pass in two direction
- Rectus femoris M



Bipennate

3- multipennate:

- Muscle fibers take more than two directions
- Deltoid M



Multipennate

Classification of muscle according to

1- arrangement of muscle fiber

Parallel (fusiform) – pennate- circular

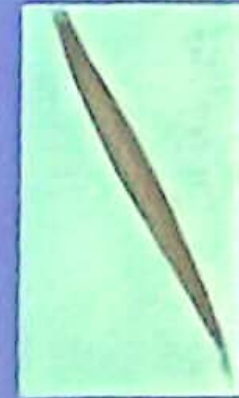
A- Parallel fibers:

1.Strap (sartorius)

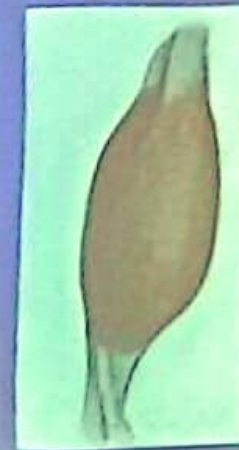
2.Quadrates (abdominal muscle)

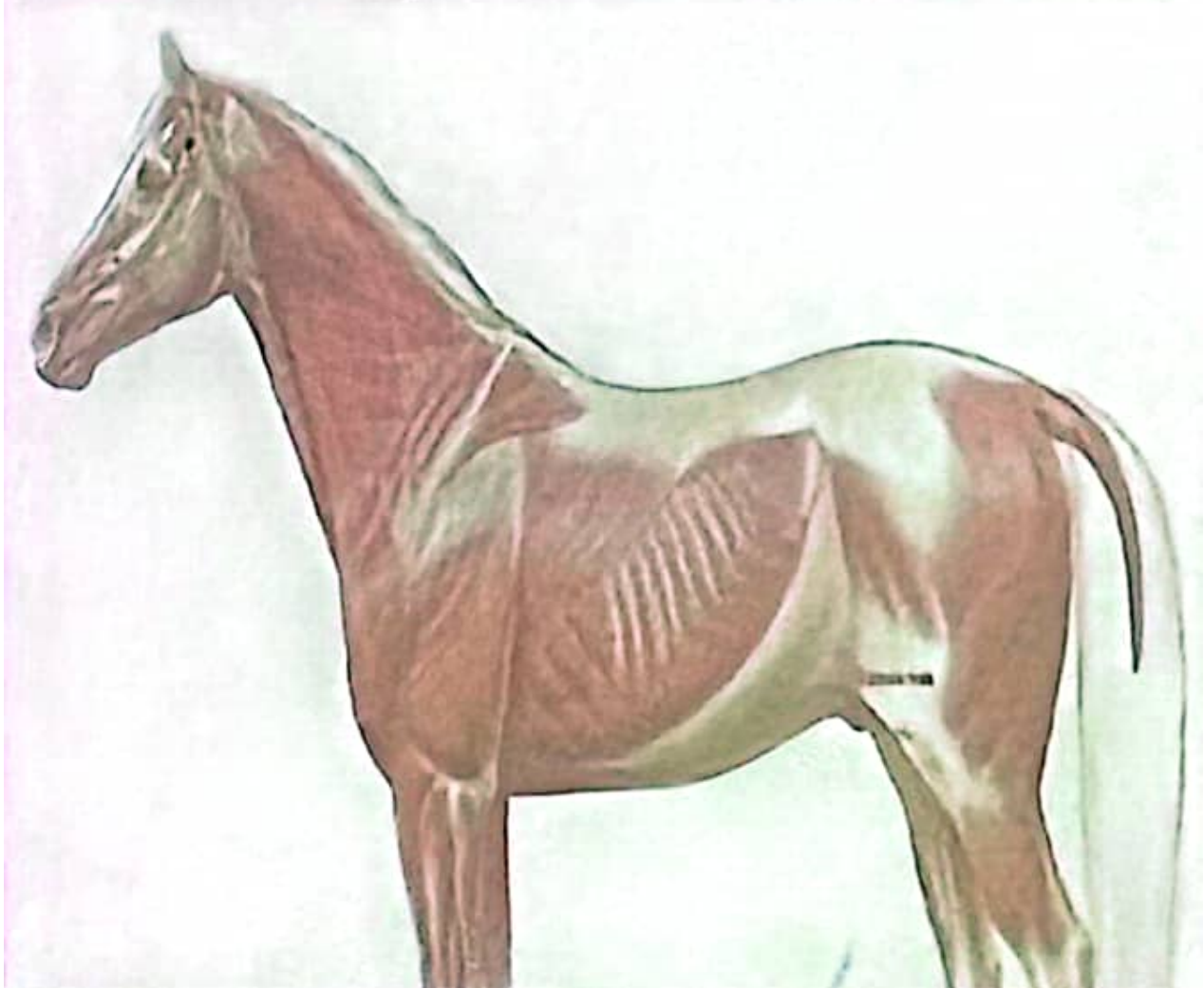
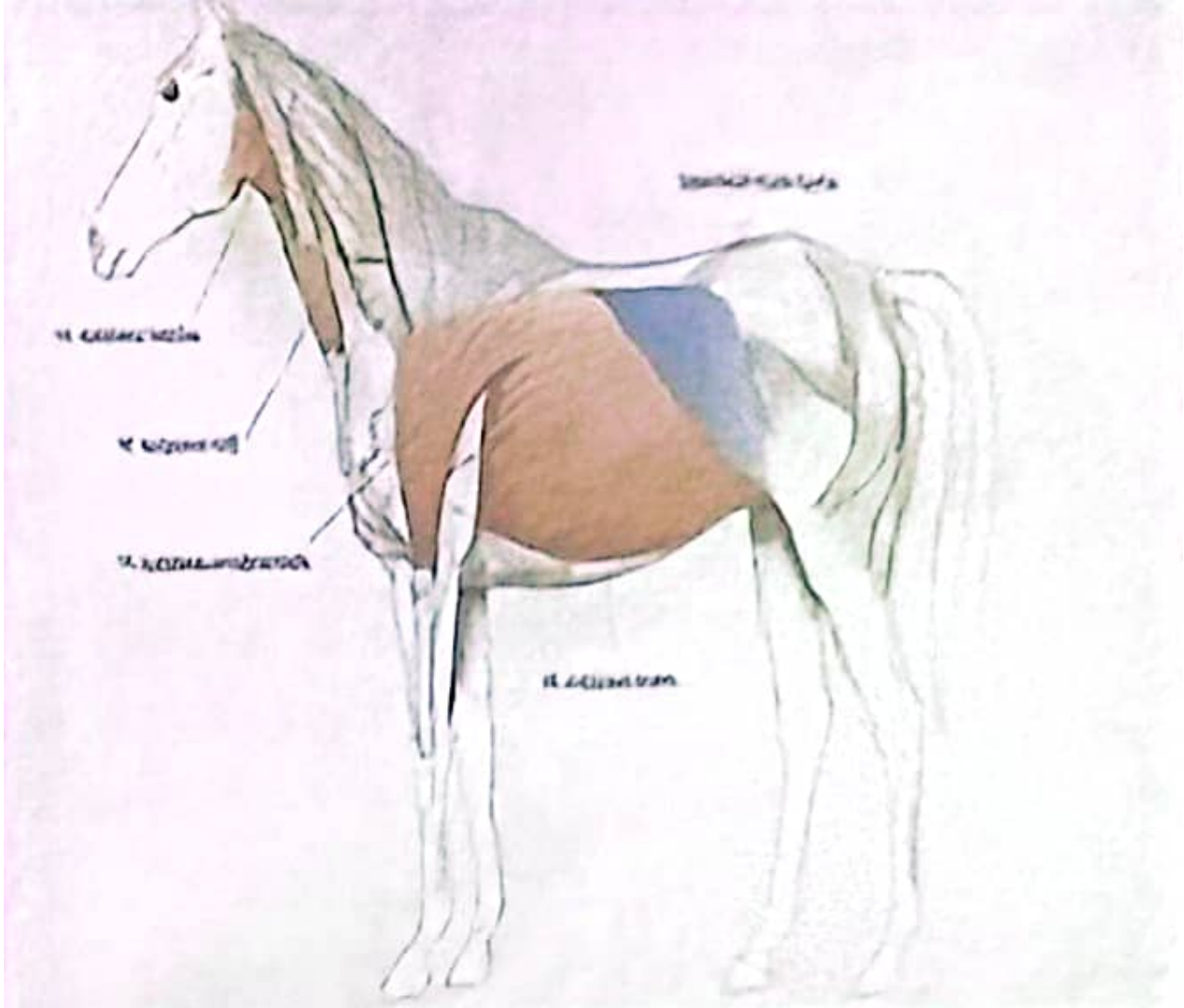
3.Fusiform (biceps brachi

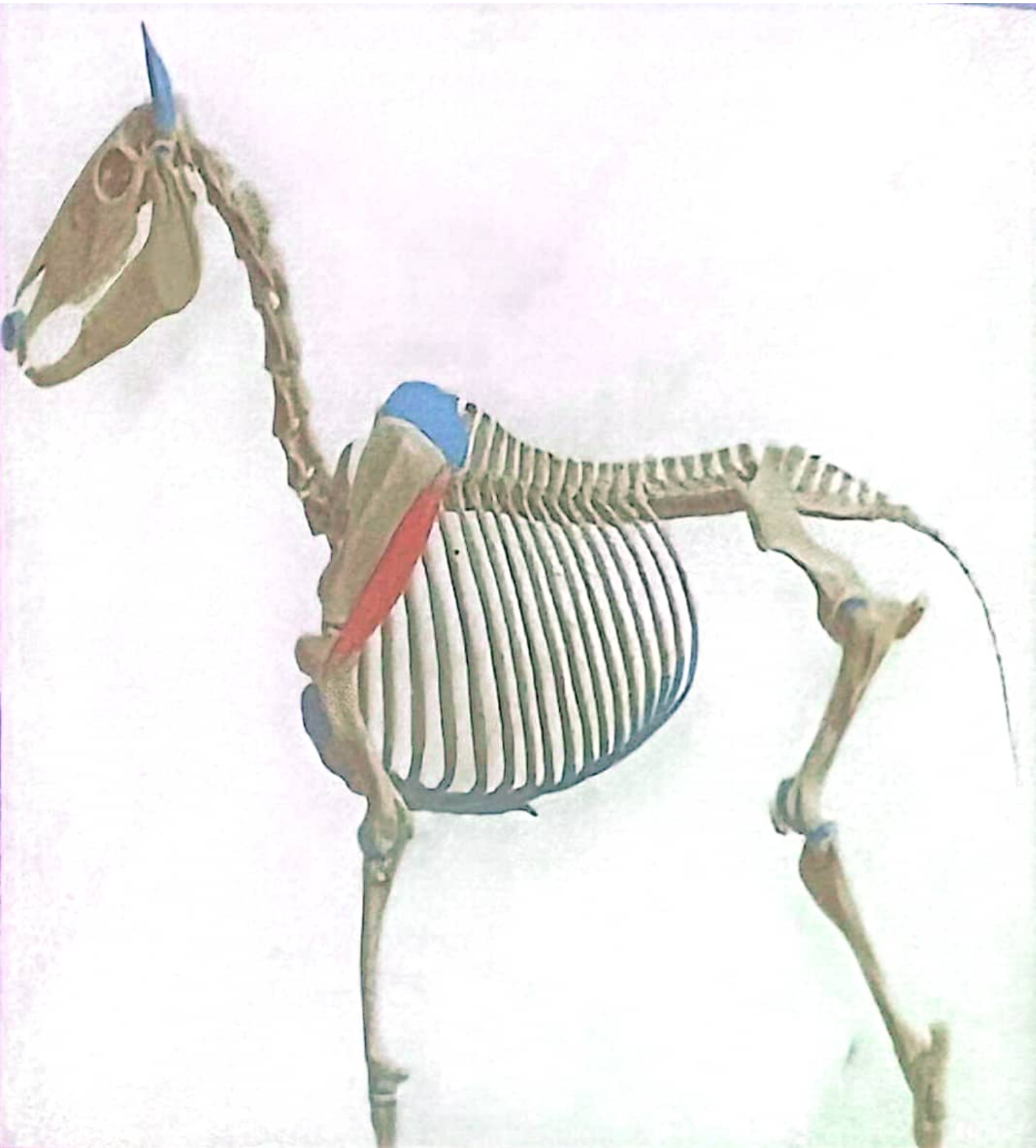
Strap



Fusiform







1. Fascia: it is a sheath of dense or loose connective tissue investing the muscles and guide in its function.

1. Superficial fascia

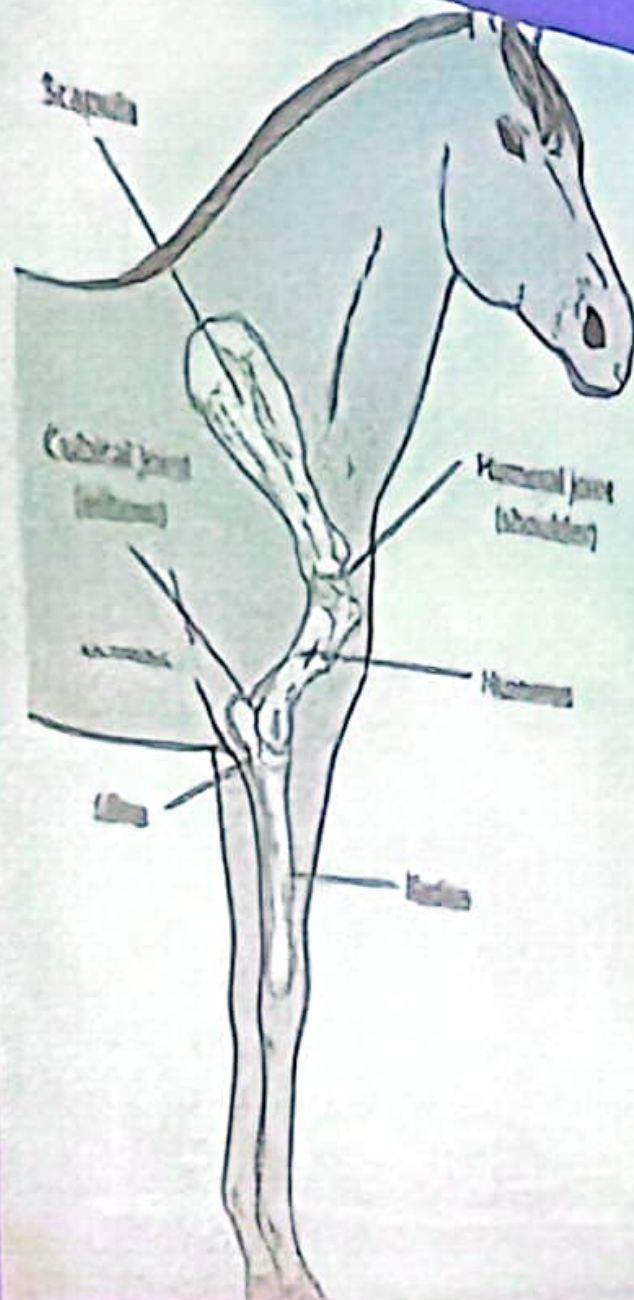
connective tissue between skin and muscles contain fat. It has cutaneous muscles in some areas move skin when a fly rests on animal skin.

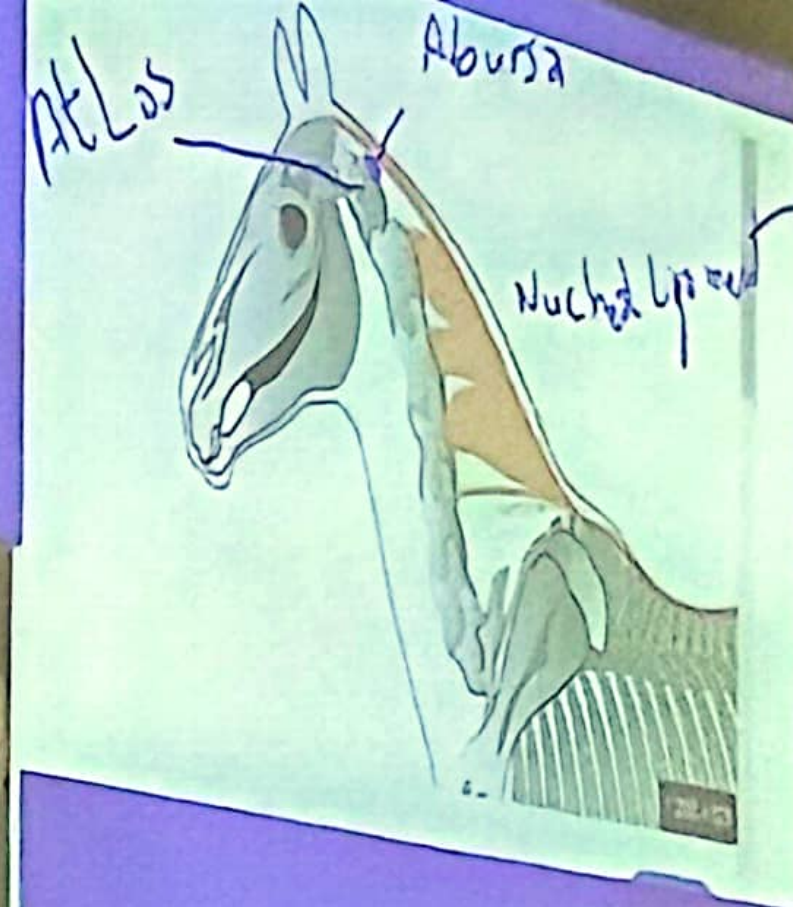
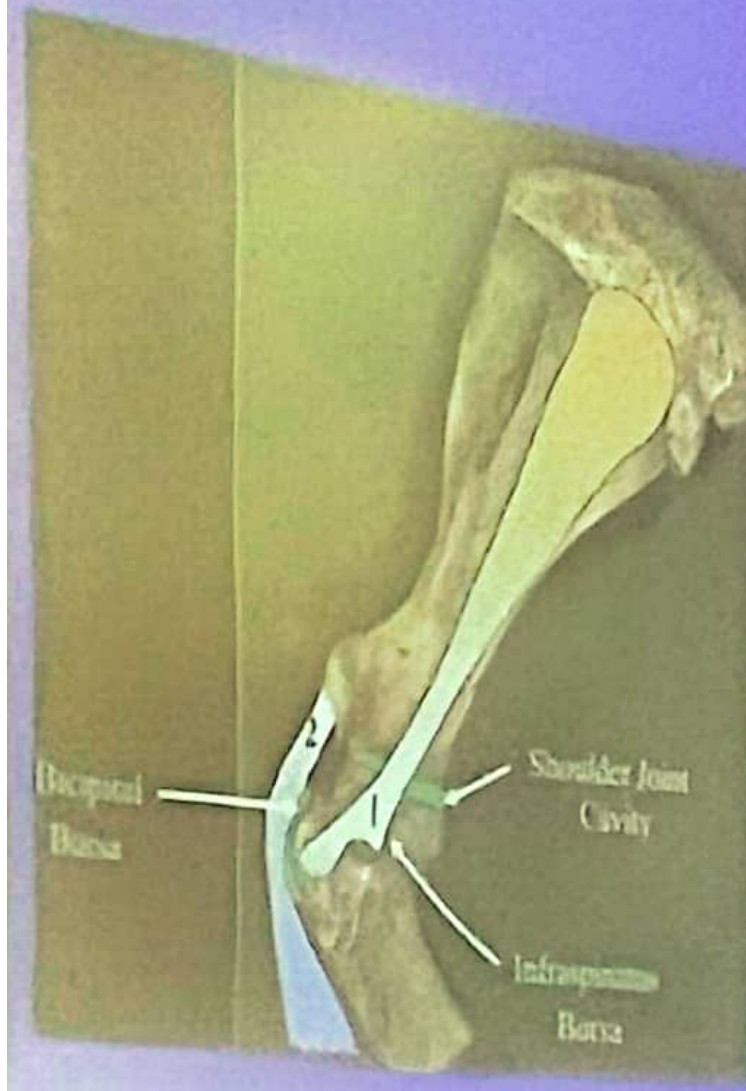
2. Deep fascia

It covers and in between the skeletal muscles.

3. Visceral fascia

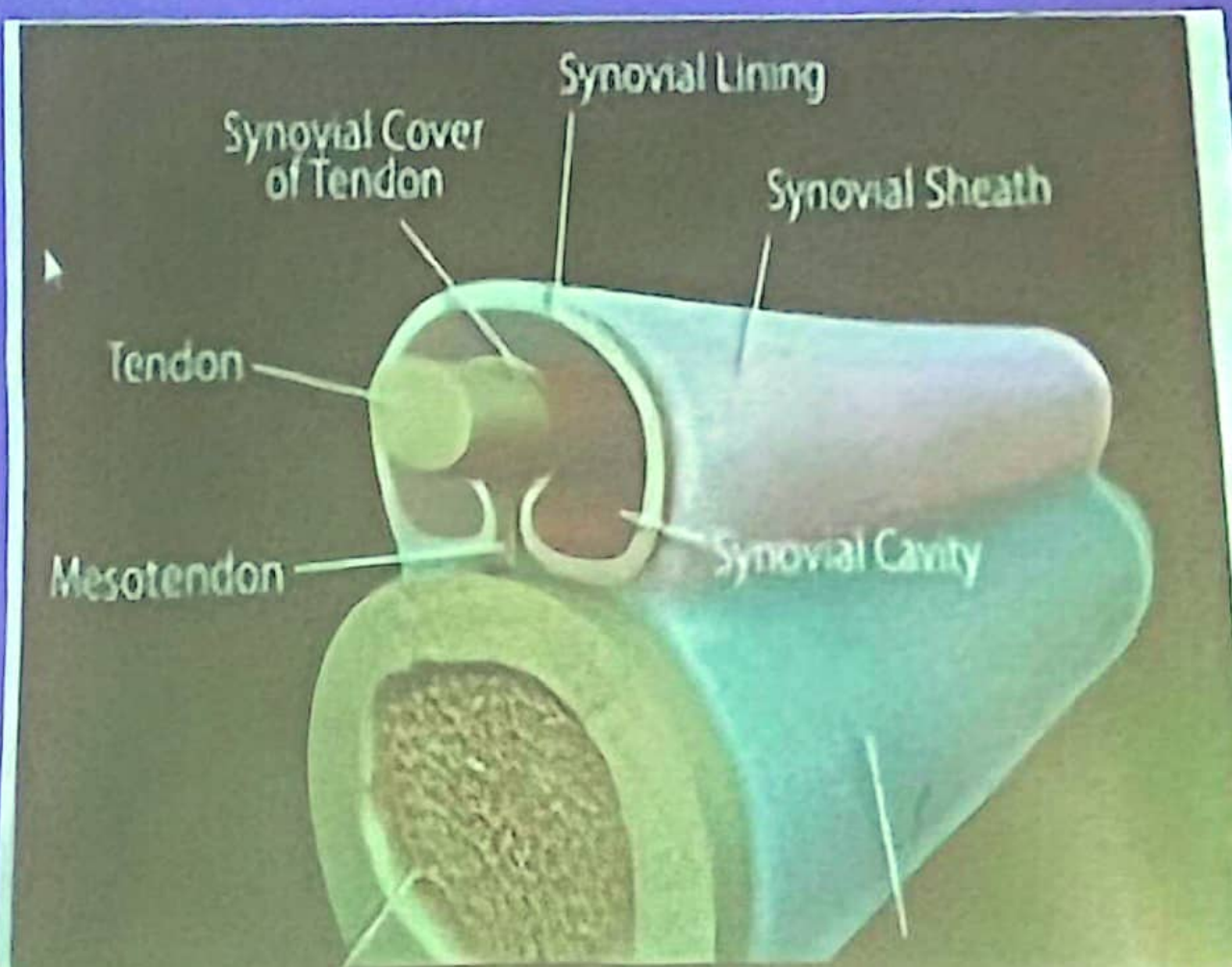
lined walls of body cavities.





2. Tendon synovial sheath

- an elongated bursa containing synovia rolled on the tendon along its length over bones to reduce the friction between tendon and bones.
- It consists of two layers closed sac around the tendon, its edges meet in mesotendon where vessels & nerves reach tendon.



Types of bursa

1. Subtendinous bursa:
between tendons and bony prominences.
Ex. bicipital bursa

2. Subligamentous bursa:
between ligaments and bony prominences.
Ex. atlantal bursa

3. Cutaneous bursa:
between the skin and bony prominences.
Ex. olecranon bursa

5. Retinaculum

- At the level of tarsus and the carpus the fascia provides a transverse bands around the tendons, attach to the bone and ligaments in order to fix the tendon in this position
- .As, flexor and extensor retinaculum of carpus

1. Synovial bursa

- Simple synovial sac filled with synovia
 - Located between two structures to decrease the friction between these structures
1. Subtendinous: bet tendon and bony prominence
 2. Subcutaneous: bet skin and bony prominence
 3. Subligamentous: bet ligament and bony prominence

